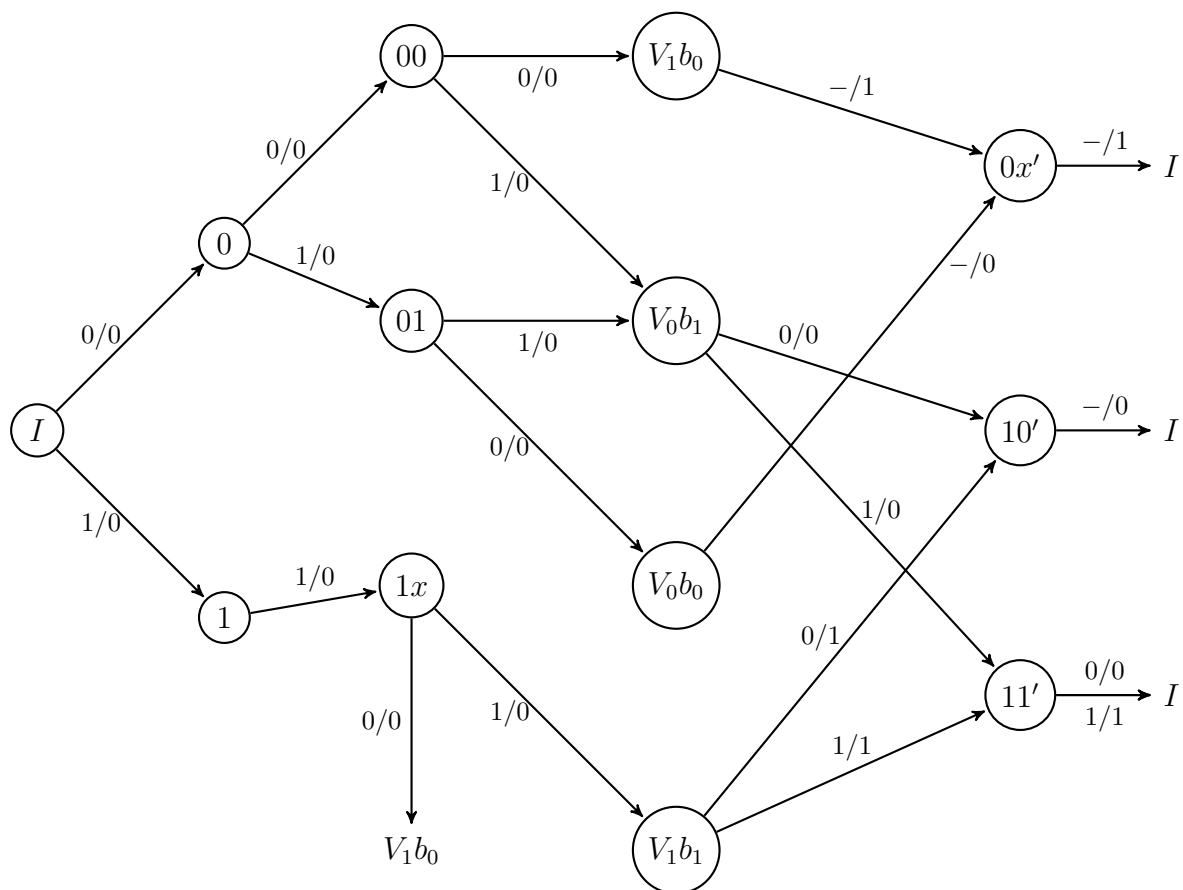


# Soluzioni

## Esercizio 1:



**Esercizio 2:**

```
 $\mu_1 : IR_x \rightarrow MAR, 0 \rightarrow B;$   
 $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$   
 $\mu_3 : MBR \rightarrow T1;$   
c: if  $OR(T1) == 1$  then  
   $\mu_2 : M[MAR] \rightarrow MBR, INCR(MAR) \rightarrow MAR;$   
  if  $MBR_{31} == 1$  then  
     $\mu_4 : MAR \rightarrow IND1;$   
     $\mu_5 : MBR \rightarrow MAR;$   
     $\mu_6 : M[MAR] \rightarrow MBR, IND1 \rightarrow MAR;$   
  else  
     $\mu_0 : \phi;$   
  end  
   $\mu_7 : MBR \rightarrow A;$   
   $\mu_8 : A + B \rightarrow B, INCR(MAR) \rightarrow MAR, DECR(T1) \rightarrow T1, goto\ c;$   
else  
   $\mu_9 : B \rightarrow AC;$   
end
```

### Esercizio 3:

---

```
%include "utils.nasm"
section .data
    V dw 15, -5, -3, -3, 6
    n equ ($-v)/2
    X dw 7

section .bss
    cnt resd 1

section .text
global _start:
extern proc
_start:
    PUSH     V
    PUSH     dword n
    PUSH     word [X]
    PUSH     cnt
    CALL     proc
    ; stampa conteggio
    printd   dword [cnt]
    exit 0
```

---

```
section .data
    v equ 18
    n equ 14
    X equ 12
    cnt equ 8

section .text
global proc
proc:
    PUSH EBP
    MOV EBP, ESP
    PUSHAD
    MOV EAX, [EBP + V]
    MOV EDI, [EBP + n]
    XOR ESI, ESI ; indice
    XOR ECX, ECX
    MOV BX, [EBP + K]
    SAL BX, 1
    INC BX ; 2*x+1
    MOV DX, [EBP + K]
    SAR DX, 1
    NEG DX ; -X/2

ciclo:
    CMP ESI, EDI ; n
    JGE esci
    CMP [EAX+ESI*2], BX
    JE avanti
    CMP [EAX+ESI*2], DX
    JE avanti
    INC ECX

avanti:
    INC ESI
    JMP ciclo

esci:
    MOV EAX, [EBP+cnt] ;salvo il valore del conteggio
    MOV [EAX], ECX
    POPAD
    POP EBP
    RET 14
```

---